Large Language Models as Virtual Survey Respondents: Evaluating Sociodemographic Response Generation

Anonymous ACL submission

Abstract

Questionnaire-based surveys are foundational to social science research and public policymaking, yet traditional survey methods remain costly, time-consuming, and often limited in scale. This paper explores a new paradigm: simulating virtual survey respondents using Large Language Models (LLMs). We introduce two novel simulation settings, namely Partial Attribute Simulation (PAS) and Full Attribute Simulation (FAS), to systematically evaluate the ability of LLMs to generate accurate and demographically coherent responses. In PAS, the model predicts missing attributes based on partial respondent profiles, whereas FAS involves generating complete synthetic datasets under both zero-context and context-enhanced conditions. We curate a comprehensive benchmark suite, **LLM-S^3** (**L**arge **L**anguage **M**odel-based **S**ociodemographic **S**urvey **S**imulation), that spans 11 real-world public datasets across four sociological domains. Our evaluation of multiple mainstream LLMs (GPT-3.5/4 Turbo, LLaMA 3.0/3.1-8B) reveals consistent trends in prediction performance, highlights failure modes, and demonstrates how context and prompt design impact simulation fidelity. This work establishes a rigorous foundation for LLM-driven survey simulations, offering scalable and cost-effective tools for sociological research and policy evaluation. Our code and dataset are available at: https://anonymous.4open.science/r/LLM-S-Cube-Benchmark/.

1 Introduction

Questionnaire-based surveys serve as critical instruments for understanding public sentiment, tracking social dynamics, and guiding evidence-based policymaking. However, the traditional survey pipeline faces mounting challenges: high operational costs, lengthy data collection timelines, and declining response rates (Mou et al., 2024). These limitations hinder the timely delivery of insights essential for responsive policy formulation. To address these issues, we propose and systematically investigate a novel approach: leveraging Large Language Models (LLMs) to simulate virtual survey respondents.

Recent advances in LLMs have demonstrated remarkable generalization capabilities in tasks ranging from question answering to synthetic text generation (Brown et al., 2020; Grattafiori et al., 2024; Törnberg, 2023). Despite growing interest in LLM-driven simulations (Dillion et al., 2023; Kim and Lee, 2023; Jansen et al., 2023), existing research often suffers from three limitations that our work addresses: (1) narrow task coverage, where prior studies frequently focus on a limited number of survey items or demographic traits, restricting their generalizability across diverse tasks (Yang et al., 2024; Zhou et al., 2024; Sun et al., 2024; Argyle et al., 2022); (2) domain-specific constraints, as most existing works are tailored to particular sociological themes (e.g., politics or consumer behavior), which impedes comprehensive cross-domain evaluation (Chaudhary and Penn, 2024; Yu et al., 2024; Argyle et al., 2022; Törnberg et al., 2023; Cao et al., 2025); and (3) inadequate benchmarking, primarily due to reliance on synthetic or overly simplified datasets that fail to reflect the complexity or scale of real-world survey distributions (Argyle et al., 2022; Ziems et al., 2024; Ren et al., 2024; Cao et al., 2025). Our research bridges these gaps by introducing a systematic and extensible framework for evaluating LLMs in survey response generation.

To this end, we introduce two distinct and novel simulation paradigms: **Partial Attribute Simulation (PAS)**, where LLMs predict missing demographic or behavioral attributes from incomplete respondent profiles, and **Full Attribute Simulation (FAS)**, where LLMs generate synthetic datasets under varying degrees of contextual constraint. These paradigms are designed to reflect practical scenarios in applied research, from imputing nonresponse in existing surveys to simulating synthetic popula-

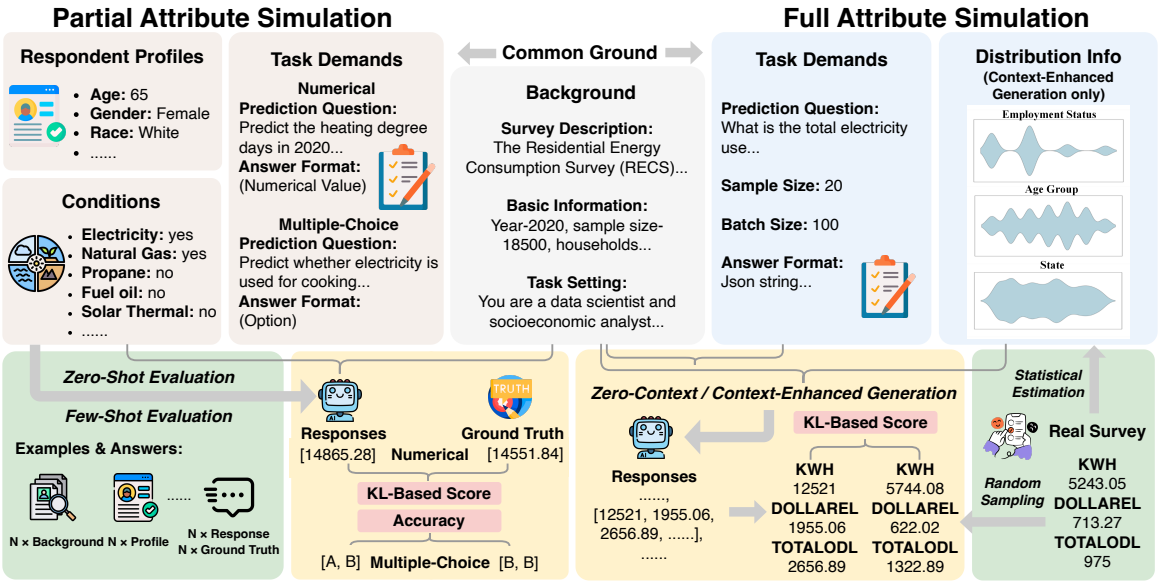


Figure 1: An Overview of Partial Attribute Simulation (left) and Full Attribute Simulation (right).

tions for complex policy experiments.

To support reproducibility and broad applicability, we curate a cross-domain benchmark suite, **LLM-S^3** (**L**arge **L**anguage **M**odel-based **S**ociodemographic **S**urvey **S**imulation), composed of eleven public datasets across four sociological areas: Social and Public Affairs, Work and Income, Household and Behavioral Patterns, and Health and Lifestyle. We evaluate four representative LLMs (GPT-3.5 Turbo, GPT-4 Turbo, LLaMA 3.0-8B, LLaMA 3.1-8B) on both numerical and categorical prediction tasks under zero-shot and few-shot settings. Our extensive experiments yield several key findings: (i) LLMs exhibit reliable and consistent performance trends across various model families and task types; (ii) prompt design and contextual augmentation significantly impact simulation fidelity, highlighting their crucial role; and (iii) failure cases in structured output generation remain a major bottleneck, particularly evident in the Full Attribute Simulation (FAS) scenario. These insights not only advance our fundamental understanding of LLMs in survey applications but also provide practical design principles for future research and deployment in this emerging field.

In sum, to the best of our knowledge, this work offers the first large-scale, systematic exploration of LLMs as virtual survey respondents, contributing both a novel simulation methodology and a robust, multi-domain benchmark to drive future advances in LLM-assisted social research.

2 Partial Attribute Simulation (PAS)

In numerous real-world survey scenarios, researchers frequently possess foundational demographic information about respondents (*e.g.*, age, gender, occupation) but lack specific insights into individual opinions or preferences on particular topics. This inherent data sparsity poses a significant challenge for accurately simulating survey responses under such constraints. To address this, we introduce the Partial Attribute Simulation (PAS) setting, a novel framework designed to evaluate LLMs' ability to predict missing sociodemographic attributes. As illustrated in Figure 1, PAS operates by providing LLMs with only fundamental demographic data and selected conditional attributes, without access to an individual's complete historical response profile. The LLMs are then tasked with generating plausible responses, effectively simulating how individuals with specific partial characteristics might answer survey questions. This paradigm offers valuable insights into the influence of partial information on survey responses, enabling the simulation of realistic survey scenarios across diverse sociological contexts.

2.1 Formulation

Within the Partial Attribute Simulation (PAS) framework, survey respondents are conceptualized as virtual agents, each characterized by a predefined set of attributes. Let $\mathcal{D} = \{\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_N\}$ denote a survey dataset of N samples, where each

$\mathbf{x}_i \in \mathbb{R}^M$ consists of M attributes from the attribute set $\mathcal{A}$. The set $\mathcal{A}$ is partitioned into two distinct subsets: the input attribute set, $\mathcal{A}_{\text{prior}}$, which are provided to the Large Language Model (LLM) via prompts, and the target attribute set, $\mathcal{A}_{\text{target}}$, that the LLM is required to infer. Formally, we define:

$$\mathcal{A}_{\text{prior}} \subseteq \mathcal{A}, \mathcal{A}_{\text{target}} = \mathcal{A} \setminus \mathcal{A}_{\text{prior}}. \quad (1)$$

Each virtual agent is then tasked with predicting the target attributes $\mathbf{A}_{\text{target}} \in \mathcal{A}_{\text{target}}$ conditioned on the given input attributes $\mathbf{A}_{\text{prior}} \in \mathcal{A}_{\text{prior}}$. This is formalized as modeling the conditional probability distribution parameterized by θ:

$$\mathbf{A}_{\text{target}} \sim P_\theta(\mathbf{A}_{\text{target}} | \mathbf{A}_{\text{prior}}). \quad (2)$$

In the PAS scenario, each respondent is simulated as an independent virtual agent using an LLM, which is required to generate a distribution over the possible values of $\mathbf{A}_{\text{target}}$ conditioned on $\mathbf{A}_{\text{prior}}$.

Numerical Prediction Tasks: When predicting numerical attributes, the primary objective is to estimate the population-level distribution of the target attribute set $\mathcal{A}_{\text{target}}$, rather than predicting specific values for individual respondents. Performance is assessed using a KL-based score (see Appendix A.2.2), a monotonic decreasing transformation of the Kullback–Leibler (KL) divergence, where higher values indicate better alignment between predicted and ground-truth distributions.

Multiple-Choice Prediction Tasks: For categorical attributes, the focus shifts from modeling distributions to determining the correct category for each individual respondent. In this context, the LLM's performance is evaluated based on the accuracy of its predictions for each respondent's target attribute, with the average accuracy across all respondents being reported. This metric directly assesses the model's classification capability.

2.2 Evaluation Strategies in PAS

In the PAS scenario, two primary evaluation strategies are employed to assess LLM performance: **Zero-Shot Evaluation** and **Few-Shot Evaluation**.

In the **Zero-Shot Evaluation**, the model generates predictions solely based on its pre-trained parametric knowledge, without any prior examples. The prompt, detailed in Figure 10 in Appendix A, is structured around five key components:

- **Background:** Consists of a survey description (purpose and structure of the questionnaire), basic information (outlining context such as year, region, domain, etc.) and a task setting description (instructions guiding the LLM to simulate a specific respondent).
- **Respondent Profiles:** Includes demographic information (*e.g.*, age, gender) to tailor predictions to individual characteristics.
- **Conditions:** Provides conditional attributes (*e.g.*, electricity usage, health factors) to guide predictions based on specific circumstances.
- **Task Demands:** Specifies the prediction task type (*e.g.*, numerical or multiple-choice).
- **Answer Format:** Defines the output format (*e.g.*, numerical/option lists, or JSON strings).

In the **Few-Shot Evaluation**, the model refines its predictions of $\mathcal{A}_{target}$ by learning from a small set of in-context examples (Dong et al., 2024). These exemplars act as demonstrations, helping the model recognize the expected response format and improve predictive accuracy without explicit task-specific training. In this study, the LLM receives five illustrative examples, providing structural guidance and reinforcing expected patterns. See Figure 10 in Appendix A for a prompt example.

Experimental Setup: Based on these two evaluation strategies, we evaluated public datasets from four domains (see Table 4) with key attributes selected for prediction (see Table 5). Numerical attributes were evaluated using a KL-based performance score (refer to Appendix A.2.2 for calculation details), while multiple-choice tasks were assessed by accuracy. Performance across both tasks, evaluated using OpenAI's GPT-3.5 Turbo, GPT-4 Turbo (Brown et al., 2020), and Meta's LLaMA 3.0 and 3.1 (8B) models (Grattafiori et al., 2024), is presented in Table 1 and 2. GPT models were accessed via their official API[1], and LLaMA models were deployed locally using the Ollama platform[2] on an NVIDIA A6000 GPU, with default hyperparameters applied for unbiased comparisons.

2.3 PAS Performance Analysis

To better understand how model series and evaluation strategies affect performance across different task types, we examine aggregate trends in the PAS scenario. As shown in Figure 2, the GPT series generally outperforms LLaMA on multiple-choice tasks (46.82% *vs.* 43.04%), while LLaMA achieves slightly better results on numerical prediction (0.5255 *vs.* 0.5194). Few-shot evaluation

[1] https://openai.com/api/
[2] https://ollama.ai/

Table 1: LLM performance on numerical prediction in PAS. The KL-based scores (↑) are reported as *avg.* ± *std.*: bold for the best and underlined for the second-best.

Dataset	Zero-Shot Evaluation				Few-Shot Evaluation			
	GPT-3.5 Turbo	GPT-4 Turbo	LLaMA 3.0	LLaMA 3.1	GPT-3.5 Turbo	GPT-4 Turbo	LLaMA 3.0	LLaMA 3.1
ANES	**0.5735**$_{\pm 0.0529}$	0.5576$_{\pm 0.0293}$	0.4875$_{\pm 0.0533}$	0.4768$_{\pm 0.0755}$	**0.5568**$_{\pm 0.0072}$	0.5460$_{\pm 0.0494}$	0.4646$_{\pm 0.0575}$	0.4683$_{\pm 0.0334}$
GSS	0.6983$_{\pm 0.0120}$	**0.7082**$_{\pm 0.0129}$	0.6990$_{\pm 0.0198}$	0.6818$_{\pm 0.0062}$	0.6766$_{\pm 0.0086}$	**0.7054**$_{\pm 0.0164}$	0.6879$_{\pm 0.0142}$	0.6800$_{\pm 0.0308}$
BIS	**0.6204**$_{\pm 0.0462}$	0.5119$_{\pm 0.0129}$	0.5777$_{\pm 0.0458}$	0.5730$_{\pm 0.0378}$	**0.6207**$_{\pm 0.0417}$	0.5299$_{\pm 0.0337}$	0.5640$_{\pm 0.0486}$	0.5622$_{\pm 0.0476}$
RECS	**0.6554**$_{\pm 0.0162}$	0.5648$_{\pm 0.0568}$	0.3114$_{\pm 0.0801}$	0.2669$_{\pm 0.0197}$	0.5293$_{\pm 0.1865}$	**0.5489**$_{\pm 0.0920}$	0.3088$_{\pm 0.0874}$	0.2365$_{\pm 0.0226}$
ACS	0.5159$_{\pm 0.1283}$	0.4722$_{\pm 0.1859}$	**0.6056**$_{\pm 0.0983}$	0.5766$_{\pm 0.1239}$	0.4791$_{\pm 0.1106}$	0.3636$_{\pm 0.0410}$	**0.6054**$_{\pm 0.1047}$	0.5969$_{\pm 0.1112}$
Trell SMU	0.3523$_{\pm 0.0340}$	0.4301$_{\pm 0.0033}$	**0.5612**$_{\pm 0.0003}$	0.5361$_{\pm 0.0064}$	0.4274$_{\pm 0.0499}$	0.5197$_{\pm 0.0670}$	**0.5462**$_{\pm 0.0189}$	0.4685$_{\pm 0.0147}$
NHTS	0.4010$_{\pm 0.0066}$	0.3125$_{\pm 0.0842}$	0.4927$_{\pm 0.1769}$	**0.6079**$_{\pm 0.1036}$	0.3568$_{\pm 0.0997}$	0.3102$_{\pm 0.1499}$	0.4908$_{\pm 0.1058}$	**0.5799**$_{\pm 0.0760}$

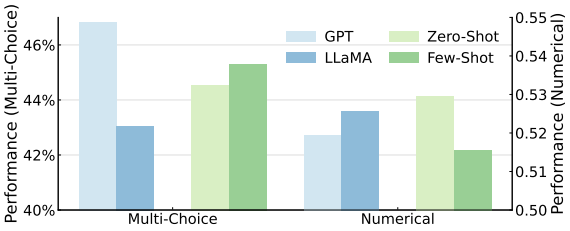


Figure 2: Average PAS performance (↑) across two tasks, grouped by model series and evaluation strategy.

tends to improve performance on multiple-choice tasks (45.32% *vs.* 44.54%), whereas zero-shot proves more effective for numerical tasks (0.5296 *vs.* 0.5154). These divergent patterns highlight the need for a more granular, task-specific analysis.

Numerical Prediction Tasks: Table 1 reveals that GPT models achieve the best performance on four datasets, while LLaMA leads on three. Although model series exhibit consistent performance trends, the effect of evaluation strategy is more variable. Few-shot settings improve performance in some cases (*e.g.*, GPT-4 Turbo on Trell SMU), but degrade it in others (*e.g.*, on ACS), underscoring its sensitivity to dataset properties and prompt design (as further discussed in Section 4.3).

Multiple-Choice Prediction Tasks: As presented in Table 2, the majority of models outperform the random-choice accuracy, with particularly pronounced improvements observed on the NHTS and Trell SMU datasets. GPT models, particularly GPT-4 Turbo, generally attain the highest accuracy across most datasets, with LLaMA outperforming them on a few datasets. Consistent with numerical tasks, our few-shot evaluation exhibits variable improvements in this prediction task as well.

3 Full Attribute Simulation (FAS)

Large-scale sociological analysis frequently necessitates understanding population-level trends and aggregate distributions, extending beyond individual-level responses. While the PAS scenario addresses the prediction of specific attributes for individual agents, the Full Attribute Simulation (FAS) tackles a distinct and more complex challenge: generating comprehensive synthetic datasets that accurately preserve underlying demographic distributions while inferring unobserved attributes. As illustrated in Figure 1, this capability is particularly critical for policy formulation when only aggregated statistics (*e.g.*, age/gender distributions) are available, but granular, domain-specific insights (*e.g.*, opinions on urban development) remain unobserved. FAS presents a compelling paradigm for synthesizing complete and statistically representative populations, offering a robust foundation for sophisticated analytical tasks in social science research and evidence-based policy design.

3.1 Formulation

The Full Attribute Simulation (FAS) formalizes the generation of multi-attribute survey data samples through two complementary paradigms, each specifically addressing distinct methodological requirements. Let $\mathcal{D}_{\text{syn}} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}$ denote the synthetic dataset containing N instances, where each $\mathbf{x}_i \in \mathbb{R}^M$ comprises M attributes spanning demographic, behavioral, and contextual dimensions. FAS aims to approximate the joint distribution $P(\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_M)$ adhering to predefined constraints, implemented via two operational modes: **zero-context** and **context-enhanced** generation.

Zero-Context Generation: The zero-context generation operates under the assumption that LLMs can probabilistically reconstruct plausible multivariate distributions solely from their parametric knowledge (Cao et al., 2025; Anthis et al., 2025). Formally, the synthetic instances are sampled as:

$$\mathbf{x} \sim P_\theta(\mathbf{X}|\mathbf{C}, \mathbf{S}), \tag{3}$$

Table 2: LLM performance on multiple-choice prediction in PAS. Accuracy (↑) is reported as avg. ± std.: bold for the best and underlined for the second-best. Cell color reflects relative performance change compared to random-choice accuracy: greener for greater improvement and redder for greater degradation.

Dataset	Zero-Shot Evaluation				Few-Shot Evaluation			
	GPT-3.5 Turbo	GPT-4 Turbo	LLaMA 3.0	LLaMA 3.1	GPT-3.5 Turbo	GPT-4 Turbo	LLaMA 3.0	LLaMA 3.1
ANES	0.7854$_{\pm 0.0598}$	**0.8358**$_{\pm 0.0324}$	0.7827$_{\pm 0.0487}$	0.7525$_{\pm 0.0372}$	0.7796$_{\pm 0.0447}$	**0.8219**$_{\pm 0.0395}$	0.7686$_{\pm 0.0382}$	0.7355$_{\pm 0.0491}$
GSS	0.5588$_{\pm 0.1359}$	**0.6313**$_{\pm 0.0340}$	0.5289$_{\pm 0.0583}$	0.4969$_{\pm 0.0326}$	0.4948$_{\pm 0.0430}$	**0.6223**$_{\pm 0.0349}$	0.4933$_{\pm 0.0895}$	0.4129$_{\pm 0.0778}$
BIS	0.1865$_{\pm 0.0032}$	0.2165$_{\pm 0.0022}$	0.2629$_{\pm 0.0600}$	**0.2848**$_{\pm 0.0217}$	0.1465$_{\pm 0.0164}$	0.2169$_{\pm 0.0045}$	**0.2261**$_{\pm 0.0233}$	0.2077$_{\pm 0.0044}$
RECS	0.6208$_{\pm 0.1116}$	**0.8598**$_{\pm 0.0793}$	0.7257$_{\pm 0.0144}$	0.6540$_{\pm 0.0730}$	0.6386$_{\pm 0.0257}$	**0.9113**$_{\pm 0.0165}$	0.7483$_{\pm 0.2132}$	0.6742$_{\pm 0.0623}$
EmpS	0.2191$_{\pm 0.0293}$	0.2171$_{\pm 0.0017}$	**0.2302**$_{\pm 0.0259}$	0.2188$_{\pm 0.0183}$	**0.2267**$_{\pm 0.0092}$	0.2116$_{\pm 0.0105}$	0.2147$_{\pm 0.0230}$	0.2143$_{\pm 0.0045}$
NHTS	**0.7314**$_{\pm 0.0077}$	0.7231$_{\pm 0.0003}$	0.5732$_{\pm 0.0126}$	0.6507$_{\pm 0.0453}$	0.7246$_{\pm 0.0322}$	**0.7300**$_{\pm 0.0057}$	0.6409$_{\pm 0.0797}$	0.5987$_{\pm 0.0270}$
Trell SMU	0.6415$_{\pm 0.0105}$	**0.6525**$_{\pm 0.0050}$	0.3865$_{\pm 0.0138}$	0.5028$_{\pm 0.0134}$	0.7228$_{\pm 0.0357}$	0.6601$_{\pm 0.0057}$	0.7099$_{\pm 0.0051}$	**0.7309**$_{\pm 0.0176}$
MxMH	**0.1283**$_{\pm 0.0380}$	0.1021$_{\pm 0.0018}$	0.0901$_{\pm 0.0129}$	0.0933$_{\pm 0.0151}$	**0.1542**$_{\pm 0.0517}$	0.0977$_{\pm 0.0093}$	0.0920$_{\pm 0.0103}$	0.0860$_{\pm 0.0120}$
YPS	0.2729$_{\pm 0.0074}$	0.2559$_{\pm 0.0038}$	**0.3083**$_{\pm 0.0479}$	0.2787$_{\pm 0.0426}$	0.2916$_{\pm 0.0283}$	**0.2968**$_{\pm 0.0471}$	0.2768$_{\pm 0.0175}$	0.1990$_{\pm 0.0418}$
MHD	0.3377$_{\pm 0.0058}$	0.3375$_{\pm 0.0045}$	0.3390$_{\pm 0.0068}$	**0.3434**$_{\pm 0.0070}$	0.3314$_{\pm 0.0198}$	0.3339$_{\pm 0.0079}$	0.3403$_{\pm 0.0094}$	**0.3437**$_{\pm 0.0063}$

where $\mathbf{C}$ encapsulates contextual metadata (*e.g.*, "U.S. households in 2022"), $\mathbf{S}$ defines schema constraints (*e.g.*, attribute types, value ranges), and θ represents the LLM's parameters encoding implicit sociodemographic patterns. This paradigm eliminates external distributional guidance, relying entirely on the model's emergent understanding of attribute correlations. For instance, when simulating energy consumption (RECS dataset), the schema specifies permissible ranges for *annual kWh usage* $(500–15,000)$ and *employment status* (categorical options: employed/retired), while the LLM autonomously infers relationships between income brackets and electricity expenditure.

Context-Enhanced Generation: In contrast, context-enhanced generation explicitly conditions the synthesis process on empirical priors, such as population-level statistics or attribute distributions, to better align synthetic generations with real-world patterns. Synthetic instances are sampled as:

$$\mathbf{x} \sim P_\theta(\mathbf{X}|\mathbf{C}, \mathbf{S}, P(\mathbf{A}_{\text{prior}})), \tag{4}$$

where $P(\mathbf{A}_{\text{prior}})$ injects known marginal or conditional distributions (*e.g.*, age-gender cohorts by state, education-income correlations). This integration occurs through structured prompting. For example, appending U.S. Census-derived age distributions ($\mu = 45.2$, $\sigma = 18.3$) and party affiliation rates (Democrat: 43%, Republican: 37%) when simulating political surveys (ANES). LLMs then perform constrained sampling, iteratively refining proposals to align with provided prior distributions, akin to a learned rejection sampling algorithm.

To validate the performance of LLMs in the FAS scenario under the two generation paradigms, we select four representative datasets, maintaining the same LLMs and experimental conditions used in the PAS scenario. The evaluation is performed using the KL-based performance score (see Appendix A.2.2), which assesses the alignment between the generated and ground-truth distributions.

3.2 FAS Performance Analysis

Table 3 presents the FAS performance of various LLMs, providing valuable insights into their ability to adapt to different generation paradigms, especially when additional context is provided. In general, GPT-4 Turbo demonstrates superior performance across all datasets and generation paradigms. This overall superiority is likely due to the larger parameter scale of GPT models, which enables them to better handle complex reasoning and inference tasks. The increased scale allows GPT-4 Turbo to process and integrate additional context more effectively than the 8B versions of LLaMA models selected here, contributing to its superior performance in the FAS scenario. In addition, we observe that the top-performing and second-best LLMs (marked with bold and underlined) generally belong to the same LLM series, demonstrating a clear pattern of consistency across datasets.

On the other hand, the context-enhanced generation generally achieves better results than the zero-context generation, with most models showing performance improvements when given additional context. Notably, GPT-4 Turbo demonstrates greater contextual utilization than its GPT-3.5 counterpart, achieving larger performance gains across datasets. This implies that integrated context helps models refine decisions through latent knowledge activation for the FAS scenario. Nevertheless, some models exhibit slight performance declines (*e.g.*, GPT-3.5 Turbo on RECS; LLaMA 3.1 on Trell

Table 3: LLM performance in FAS. The KL-based scores (↑) are reported as *avg.* ± *std.*: bold for the best and underlined for the second-best. Color percentage indicates relative performance change from zero-context to context-enhanced generation: green for improvement and red for degradation.

Dataset	Zero-Context Generation				Context-Enhanced Generation			
	GPT-3.5 Turbo	GPT-4 Turbo	LLaMA 3.0	LLaMA 3.1	GPT-3.5 Turbo	GPT-4 Turbo	LLaMA 3.0	LLaMA 3.1
GSS	0.6490	**0.6490**	0.6508	0.6520	0.6569 ↑1.22%	**0.6862** ↑5.73%	0.6571 ↑0.98%	0.6726 ↑3.16%
Trell SMU	0.3170	**0.3320**	0.2353	0.2992	0.3410 ↑7.57%	**0.3617** ↑8.95%	0.2557 ↑8.68%	0.2792 ↓6.67%
YPS	0.6674	**0.7008**	0.6567	0.6549	0.6764 ↑1.35%	**0.7210** ↑2.89%	0.6569 ↑0.04%	0.6539 ↓0.16%
RECS	0.6334	**0.6384**	0.5597	0.3826	0.6092 ↓3.81%	**0.6403** ↑0.30%	0.5883 ↑5.12%	0.5529 ↑44.51%

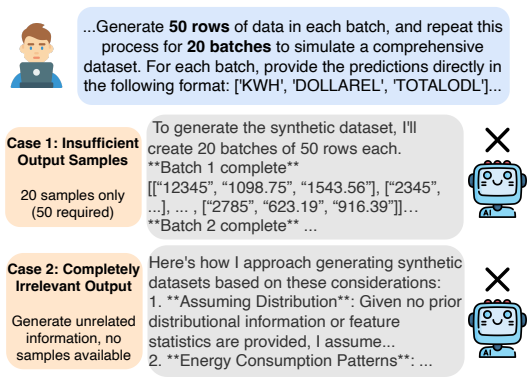


Figure 3: Representative error cases in FAS.

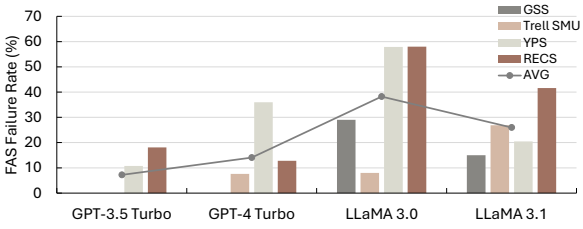


Figure 4: Statistical analysis of failure rates across different attributes in FAS.

SMU and YPS). These exceptions indicate that context effectiveness remains dataset- and model-dependent, with variations in contextual processing capabilities across architectures.

3.3 FAS Failure Phenomenon

A major challenge in the FAS scenario is the occurrence of FAS failures, where a significant proportion of generated outputs do not meet our requirements, leading to a loss of usable samples. We identify the following two failure modes, with representative examples shown in Figure 3:

- **Case 1: Insufficient Output Sample**
  In certain cases, the model fails to produce the required number of samples, leading to an incomplete representation of the underlying statistical distribution.
- **Case 2: Completely Irrelevant Output**
  The model generates non-structured outputs, such as textual descriptions, instead of the expected structured data table, making it unsuitable for quantitative analysis.

Experimental results in Figure 4 reveal significant performance disparities: LLaMA variants exhibit 15–58% failure rates across all datasets. In contrast, GPT models maintain near-zero failure rates on simpler tasks (*e.g.*, GSS and Trell SMU), but show moderate errors on more complex datasets. This divergence principally originates from two architectural advantages in the GPT series. (i) **Model Size**: their superior parametric capacity (175B *vs.* 8B) enables precise modeling of complex conditional distributions, which is a critical requirement for attribute simulation (Brown et al., 2020; Kaplan et al., 2020). (ii) **Alignment Optimization**: GPT models benefit from advanced alignment strategies such as Reinforcement Learning from Human Feedback (RLHF) (Achiam et al., 2023; Ouyang et al., 2022), which enhance their ability to follow instructions and maintain structured output fidelity. In contrast, LLaMA 3 employs a lighter alignment pipeline comprising Supervised Fine-Tuning (SFT), Direct Preference Optimization (DPO), and rejection sampling (Grattafiori et al., 2024; Rafailov et al., 2023), which improves efficiency and training scalability but may underperform on tasks requiring strict format adherence (Yao et al., 2025).

4 Discussions

Our experiments reveal critical insights into optimizing LLMs for survey simulation across three dimensions: reasoning-oriented distilled models, contextual modulation, and in-context learning dynamics. These findings provide actionable guidelines for deploying LLMs in sociological research while highlighting remaining challenges.

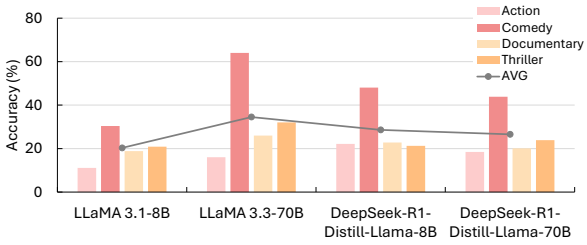


Figure 5: Performance Comparison between LLaMA models and their DeepSeek-R1-Distill counterparts.

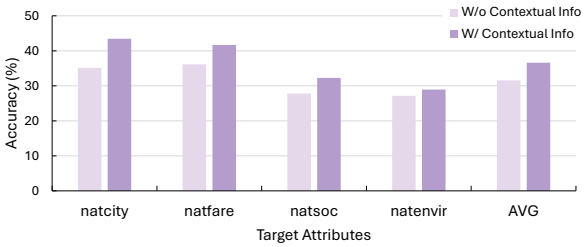


Figure 6: Impact of contextual prompts on predictive accuracy in multiple-choice tasks.

4.1 Leveraging Reasoning-Oriented Distilled Models

As survey simulation requires nontrivial reasoning over attributes, we investigate distilled LLMs explicitly optimized for reasoning, such as those from the DeepSeek-R1-Distill family (Guo et al., 2025). To evaluate their effectiveness, we compare sociodemographic response generation in the PAS scenario on the YPS dataset between standard LLaMA models (3.1-8B and 3.3-70B) and their reasoning-optimized distilled counterparts (DeepSeek-R1-Distill-Llama-8B and -70B). Figure 5 reveals that the reasoning-oriented distillation leads to a notable performance improvement for the 8B LLaMA model (↑40.6% relatively), indicating enhanced response generation capabilities. However, for the 70B LLaMA model, the distillation does not provide a significant advantage, with performance decreasing (↓22.9% relatively).

These findings suggest that while the reasoning-oriented distillation effectively boosts the performance of smaller models, its benefits diminish in larger models under the PAS setting. One possible explanation is that larger models, with their extensive parameter space, already possess the capacity to generate high-quality responses without requiring distillation (Kaplan et al., 2020), and applying further reasoning-oriented distillation may induce excessive internal deliberation in complex tasks, potentially impairing output accuracy or coherence (Cuadron et al., 2025). In contrast, smaller models, limited by fewer parameters, benefit more from the distillation as it helps streamline their processing and enhances their ability to simulate accurate sociodemographic responses (Ramesh et al., 2025).

4.2 Context-Aware Prediction Enhancement

To demonstrate the impact of integrating prompt-based contextual information, we show that embedding precise contextual cues, such as explicit references to domains, timeframes, or real-world policy frameworks, significantly enhances prediction accuracy in sociodemographic response generation. Our experiments with the LLaMA 3.1-8B model reveal that the contextual information enables the model to activate latent knowledge and produce predictions that are more closely aligned with empirical scenarios. To evaluate the effectiveness of context-aware prediction, we concentrated on four key domains of public opinion concerning government spending in the GSS dataset. Contextual information was introduced to guide the model's response generation. For example, the prompt "*Considering the environmental policies of the United States in 2022*" was used to provide contextual grounding in the environmental domain.

Incorporating contextual prompts led to improved prediction performance across the four domains, namely environmental protection (*natenvir*), big city issues (*natcity*), welfare programs (*natfare*), and social security (*natsoc*), with relative gains ranging from 6.67% to 23.62% (see Figure 6). These results establish context integration as a critical requirement for survey simulation systems, governed by two synergistic mechanisms: **first**, contextual anchoring overrides LLMs' tendency for shallow pattern matching when addressing complex societal issues; **second**, domain-specific knowledge activation closes the abstraction gap between model outputs and tangible policy dynamics (Sahoo et al., 2024). Crucially, these findings mandate a paradigm shift reframing context-aware scaffolding not as an optional add-on, but as a necessary component to be systematically implemented in high-stakes applications. This shift ensures that LLMs evolve from mere statistical pattern extractors into contextually grounded reasoning systems capable of supporting societal decision modeling.

4.3 Mitigating In-Context Learning Bias

Prior research demonstrates that few-shot learning exhibits predictive bias under class-imbalanced ex-

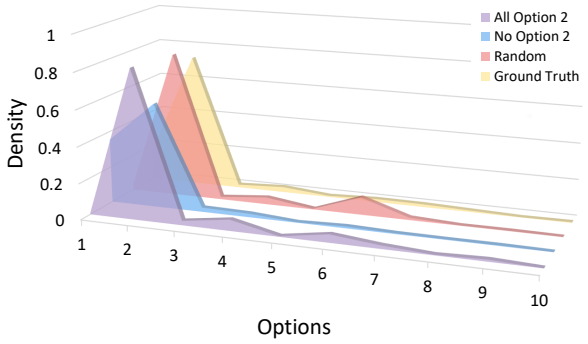


Figure 7: Comparison of output distributions under different few-shot example configurations.

ample distributions (Ochal et al., 2023), advocating balanced sampling to mitigate bias and enhance modeling validity. Building on this insight, we examine our setting focused on respondents' travel mode on travel days (i.e., *TRPTRANS*) in the NHTS dataset, and observe significant accuracy fluctuations driven by different example distributions.

We manipulated example distributions using three configurations and analyzed their output distributions, highlighting the ratio of option 2: (i) **all option 2** (82.38%), (ii) **no option 2** (57.76%), and (iii) **random selection** (80.01%). Notably, configuration (ii) caused a 24.62 percentage point decline compared to (i) (see Figure 7), underscoring the strong influence of embedded input distributions on model outputs. Statistical evidence from our experiments reveals a strong dependency between example balance and predictive robustness in our few-shot evaluation, directly motivating our adoption of strict example balancing measures. Crucially, these findings ground our operational decision to enforce balanced few-shot demonstrations in standardized prompts, ensuring counterfeit-resistant predictions aligned with real-world policy dynamics.

5 Related Work

5.1 LLM Role-Playing and Personalization

Recent advancements in LLMs have improved the realism and contextual relevance of agent behaviors across domains. A key focus has been on persona-driven role-playing, where maintaining coherent persona traits and consistent decision-making is essential (Peng and Shang, 2024; Wang et al., 2024; Xu et al., 2024). To enhance personalization further, user-specific information such as traits and profiles has been incorporated into dialogue agents, enabling responses that better reflect individual users (Zhang et al., 2018; Salemi et al., 2024). However, integrating such personalized data with large language models remains challenging, which highlights ongoing opportunities for improving adaptability (Chen et al., 2024). Complementary efforts have developed evaluation frameworks such as ECHO to measure how well LLMs replicate human-like personalized responses, thereby advancing the authenticity and naturalness of agent interactions (Ng et al., 2024; Lotfi et al., 2024).

5.2 Using LLMs for Sociological Simulation

LLMs have shown promise in simulating complex social behaviors and demographic patterns, especially in public opinion dynamics and demographic predictions. For example, prior work has simulated large-scale opinion dynamics through modeling interactions among over 1,000 virtual citizens (Park et al., 2024) and applied multi-agent reasoning for election predictions (Yu et al., 2024). In parallel, other methods employ demographic-conditioned sampling to simulate subgroup opinions; however, validation against longitudinal survey data remains limited (Argyle et al., 2022). Some studies focus on simulating group-level statistics using conditional sampling techniques, while others predict individual-level attributes such as emotions or personality traits based on psychological metrics (Huang et al., 2023; Pan and Zeng, 2023; Xie et al., 2025). Applications also include inferring psychological dispositions from textual traces and predicting social attributes through prompt engineering (Chaudhary and Penn, 2024; Zhou et al., 2024; Egami et al., 2024; Rosenbusch et al., 2023).

6 Conclusion

To evaluate large language models in sociodemographic response generation, we propose two paradigms: Partial Attribute Simulation (PAS) and Full Attribute Simulation (FAS), each targeting distinct levels of inference and data availability. PAS addresses individual-level attribute inference under incomplete profiles, while FAS targets population-level synthesis from limited demographic signals, supporting analyses in data-sparse settings. On this basis, we further introduce LLM-S^3, a comprehensive benchmark suite spanning eleven datasets across four domains. Our evaluation reveals systematic differences across model series and prompting strategies, establishing a foundation for rigorous LLM assessment and advancing trustworthy survey simulations in computational social science.

Limitations

Our study has several limitations. First, while we introduced various prompt-based techniques, improving performance was not the primary focus of our work. Future research could investigate more advanced methods, such as fine-tuning or retrieval-augmented generation (RAG), to further improve LLM-based sociodemographic response generation. Second, we focused on general-purpose LLMs, such as the GPT and LLaMA series. Future studies could explore state-of-the-art models with enhanced reasoning capabilities, such as DeepSeek-R1 and OpenAI's GPT-4o, which may be better suited for certain domain-specific tasks. Third, content moderation mechanisms in LLMs may inadvertently constrain output diversity or introduce bias. Future work could explore uncensored versions, particularly for applications involving sensitive or fairness-related considerations.

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A Experimental Details

A.1 Details of Survey Datasets from Different Domains

In this study, we undertake a comprehensive evaluation of large language models (LLMs) by leveraging a diverse array of public survey datasets across four distinct domains: **Social and Public Affairs**, **Work and Income**, **Household and Behavioral Patterns**, and **Health and Lifestyle**. By utilizing datasets such as the American National Election Studies (ANES), General Social Survey (GSS), and the Residential Energy Consumption Survey (RECS), we ensure a multifaceted approach that captures a wide range of sociocultural factors and individual experiences. This extensive selection of datasets not only enhances the robustness of our benchmark but also allows for a nuanced understanding of how LLMs can simulate responses reflective of varied demographic backgrounds and social contexts. Through this work, we aim to contribute valuable insights into the capabilities of LLMs as virtual survey respondents, highlighting their potential in generating contextually relevant and representative sociological data. Table 4 provides the details of each selected dataset.

A.2 Evaluation Metrics

To assess the performance of the models across different tasks, we utilize distinct evaluation metrics tailored to the nature of each task.

A.2.1 Multiple-Choice Prediction Tasks

For multiple-choice prediction tasks, each dataset comprises several discrete target attributes, each treated as a classification problem with a fixed number of answer options. Model performance score S is evaluated using standard accuracy, defined for the j-th attribute as:

$$S_j = \frac{1}{M_j} \sum_{i=1}^{M_j} \mathbb{I}(\hat{y}_i^{(j)} = y_i^{(j)}), \tag{5}$$

where M_j is the number of instances for attribute j, and $\mathbb{I}(\cdot)$ is the indicator function. This metric quantifies the proportion of correct predictions.

A.2.2 Numerical Prediction Tasks

For numerical prediction tasks, each dataset may contain multiple continuous target attributes (e.g., travel duration and distance on the travel days). The goal is to approximate the overall distribution of these attributes rather than predict individual

Table 4: Summary of the survey datasets utilized in constructing the benchmark for this study. The numbers of cases and attributes represent the total sample size and variable count in each raw dataset, respectively.

Domain	Survey Name	Abbr.	Year	Description	# Cases	# Attrs.
Social and Public Affairs	American National Election Studies	ANES	2020	Individual's views on certain social and political issues.	3080 individuals	470
	General Social Survey	GSS	2022	Individual's attitudes towards work and elections.	4150 individuals	1156
	Basic Income Survey	BIS	2016-2017 spring	A survey about Europeans' opinions toward basic income.	9649 individuals	15
Work and Income	Employee Survey	EmpS	Not specified	A survey about employee life experience and job satisfaction.	3025 individuals	23
	American Community Survey	ACS	Not specified	The ACS dataset offers a nuanced exploration of socioeconomic dynamics in the United States.	2000 individuals	13
Household and Behavioral Patterns	Residential Energy Consumption Survey	RECS	2020	Household survey, energy supplier data, and consumption/expenditure estimates.	18496 households	799
	National Household Travel Survey	NHTS	2017	Household's mobility choices and daily travel behavior.	923572 queries	229
	Trell Social Media Usage Data	Trell SMU	2020	Social media user's viewing/uploading/commenting habits towards a certain video website.	488877 individuals	26
Health and Lifestyle	Music & Mental Health Survey Results	MxMH	2022	Individual's music preferences and mental health correlations.	736 individuals	33
	Young People Survey	YPS	2013	Individual's preferences, habits, and personality traits.	1011 individuals	150
	Mental Health Dataset	MHD	2014-2016	Survey results on music taste and self-reported mental health.	292364 individuals	17

values. To evaluate how well the model's predicted distribution aligns with the ground truth, we first compute the Kullback–Leibler (KL) divergence for each target attribute:

$$D_{\mathrm{KL}}(P \parallel Q) = \sum_{i=1}^{K} P_i \log \frac{P_i}{Q_i}, \tag{6}$$

where P denotes the predicted distribution and Q denotes the ground-truth distribution. To improve interpretability and facilitate comparisons across models and datasets, we define a performance score S as the normalized reciprocal of KL divergence:

$$S_j = \frac{\ln\left(1 + \frac{1}{D_j}\right)}{1 + \ln\left(1 + \frac{1}{D_j}\right)}, \tag{7}$$

which smoothly maps the divergence value $D_j \in (0, +\infty)$ to a bounded interval $(0, 1)$, where higher values indicate better alignment between the predicted and true distributions. Lower KL divergence values yield higher scores (better performance), whereas larger values result in scores closer to 0 (poorer performance).

For both prediction tasks, the performance score is computed individually for each target attribute, and the final score reported in Table 1 and Table 2 is the average across all target attributes in a dataset:

$$S_{avg} = \frac{1}{N} \sum_{j=1}^{N} S_j, \tag{8}$$

where N is the number of target attributes and S_j denotes the performance score for the j-th attribute. A higher average score indicates better overall performance across all attributes.

A.3 Attribute Configuration & Detailed Performance

Given the extensive number of attributes present in the original surveys, we collaborated with domain experts to identify a subset of representative and vital attributes that are crucial for our tasks. This selection process ensures that the chosen attributes not only encapsulate the complexity of the respondents' experiences but also enhance the relevance and effectiveness of the generated responses. For the predictive attributes, we focused on those that accurately reflect the respondents' lifestyle habits, thereby providing deeper insights into their behaviors and preferences. By employing this targeted approach, our research aims to evaluate the performance of large language models in generating contextually appropriate and meaningful responses, ultimately advancing the methodology of virtual survey respondents across multiple sociocultural domains. Based on this attribute selection principle, we designed the following prediction tasks:

PAS: Table 5 summarizes the attribute configuration used for prediction tasks in the PAS scenario, preserving the original survey attribute names. Attributes in the "Input Attributesc̈olumn that appear

with underscores are withheld when serving as prediction targets. Italicized entries indicate grouped attribute sets not shown in full. The corresponding performance results are reported in Table 6 (for multiple-choice) and Table 7 (for numerical).

FAS: Table 8 details the attribute selection for prediction tasks in the FAS scenario. Attributes marked with an underscore in the "Input Attributes" column are treated as optional, excluded in the zero-context generation, and included in the context-enhanced generation. Italicized entries refer to descriptive textual labels rather than specific attributes. The performance corresponding to each target attribute is reported in Table 9.

To provide a clearer and more intuitive overview for comparing model performance across datasets, prediction tasks, and evaluation strategies in both the PAS and FAS scenarios, Figure 8 (for PAS) and Figure 9 (for FAS) display radar plots with axis values normalized within each dataset, facilitating clearer visualization of relative performance differences across models and tasks. Across both scenarios, prediction tasks are divided into two types: multi-choice (with several options) and numerical (involving continuous value estimation). Evaluation metrics include accuracy for multi-choice tasks and a normalized reciprocal (i.e., monotonic transformation) of KL divergence for numerical tasks. Together, these metrics provide a quantitative basis for comparing model performance across diverse tasks. For detailed definitions and computational procedures, please refer to Section A.2.

To ensure the robustness and reproducibility of our findings, each evaluation trial in the PAS scenario was conducted with three independent runs under the same conditions. The reported results present both the average values and standard deviations across repetitions, which helps assess the stability and reliability of model performance. In contrast, FAS adopts a design where each run consists of multiple prompt batches. Each batch generates several data outputs, and the aggregation of these outputs across all batches forms the basis for the final performance evaluation. This structure yields many data samples within a single run, offering sufficient variation for a stable and representative assessment. Hence, additional independent runs were unnecessary for FAS, as internal variability already offers a reliable performance basis.

A.4 Prompt Instances

A.4.1 PAS

In the PAS scenario, two evaluation strategies are employed: zero-shot evaluation and few-shot evaluation. The key distinction lies in that the few-shot setting offers several example attribute–answer pairs as demonstrations, intending to enhance the LLM's understanding and improve its predictive performance. As shown in Figure 10, the example prompt distinction highlights the shift in the amount of information provided to the model.

A.4.2 FAS

In the FAS scenario, we aim to generate group-level data that reflects the distribution of attributes within a specified population. This scenario employs two generation paradigms: zero-context and context-enhanced generation. In the zero-context generation, the model generates data without any additional context about the specific population. In contrast, the context-enhanced generation incorporates distributional information, such as the proportions of gender, age, and other attributes within the dataset, which helps the model generate more realistic and representative data. Figure 11 illustrates an example of these two generation paradigms, showing how the prompt structure changes depending on whether context is provided or not.

Table 5: Attribute selection for prediction tasks in the PAS scenario. A subset of variables was chosen as target attributes based on domain expertise, with input attributes selected to ensure scientific rigor, and original survey attribute names retained. Attributes marked with an underscore in the "Input Attributes" column are excluded when serving as targets, while italicized entries denote attribute sets not listed in detail. Prediction tasks are classified as Multi-Choice or Numerical.

Dataset	**Simulation Objectives**	**Input Attributes**	**Target Attributes**	
			Multi-Choice	**Numerical**
ANES	Participation in political activities. The level of support for the candidate.	age, home_ownership, income, vote20turnoutjob, particip_count, meeting, online, moneyorg, protest, persuade, button, ftmikepence1, ftrubio1, ftthomas1, ftfauci1, ftobama1, fttrump1, ftbiden1	2-Options: meeting, online, moneyorg	ftobama1, ftbiden1, fttrump1
GSS	The attitude towards relieving work pressure. Public opinions.	age, sex, racecen1, sibs, marital, childs, region, educ, relig, income, wrkstat, occ10, hhtype1, hompop, polviews, natenvir, natheal, natcity, natcrime, nateduc, natarms, natfare, natsoc	5-Options: strmgtsup, allorglevel	nateduc, natcrime
BIS	Vote for basic income. / Support basic income or not (reasons).	country_code, age, gender, rural, dem_education_level, dem_full_time_job, dem_has_children, question_bbi_2016wave4_basicincome_awareness / vote / effect / argumentsfor / argumentsagainst	5-Options: vote	argumentsfor
RECS	Household fuel usage.	HHSEX, HHAGE, EMPLOYHH, state_postal, EDUCATION, HOUSEHOLDER_RACE, NHSLDMEM, ATHOME, MONEYPY, UGASHERE, USEEL, USENG, USELP, USEFO, USESOLAR, USEWOOD, ALLELEC	2-Options: UGASHERE	KWH
ACS	The working hours or income.	age, race, gender, birth_qrtr, citizen, lang, edu, married, disability, employment, time_to_work	-	hrs_work, income
Trell SMU	Video watching habit. / Total number of videos watched (normalized).	age_group, gender, city-tier, avgCompletion, avgTimeSpent, avgDuration, avgComments, creations, content_views, num_of_comments, weekends_trails_watched_per_day, weekdays_trails_watched_per_day	6-Options: weekdays_trails_watched_per_day	content_views
EmpS	Levels of satisfaction with the job (5 levels).	WorkEnv, EmpID, Age, Gender, MaritalStatu, JobLevel, Dept, EmpType, CommuteMode, EduLevel, haveOT, JobSatisfaction, Experience, PhysicalActivityHours, SleepHours, CommuteDistance, NumCompanies, TeamSize, NumReports, EduLevel, TrainingHoursPerYear	5-Options: WLB, WorkEnv, Workload, Stress, JobSatisfaction	-
NHTS	Trip model, trip duration, trip distance.	HHFAMINC, HOMEOWN, URBAN, HH_CBSA, DRIVER, EDUC, R_RACE, R_SEX_IMP, WORKER, TRAVDAY, TRPTRANS, GASPRICE, TRAVDAY, TDAYDATE	10-Options: TRPTRANS	TRPMILES, TRVLCMIN
MxMH	Mental health situation.	Timestamp, Age, Fav genre, Hours per day, While working, Instrumentalist, Composer, Exploratory, Classical, Country, EDM, Folk, Gospel, Hip hop, Jazz, K pop, Latin, Lofi, Metal, Pop, R&B, Rap, Rock, Video game music	11-Options: Anxiety, Depression, Insomnia, OCD	-
YPS	Degree of interests of movies and music.	Music, Slow songs or fast songs, Dance, Folk, Country, Classical music, Musical, Pop, Rock, Metal or Hardrock, Punk, Hiphop, Rap, Reggae, Ska, Swing, Jazz, Rock n roll, Alternative, Latino, Techno, Trance, Opera, Movies, Horror, Romantic, Sci-fi, War, Fantasy/Fairy tales, Animated	5-Options: Action, Documentary, Thriller, Comedy	-
MHD	Feeling the level of repression. Coping_Struggles only yes or no selection, others are three selections.	*History of mental condition*, Timestamp, Gender, Country, Occupation, self_employed, family_history, treatment, Days_Indoors, Changes_Habits, Mental_Health_History, mental_health_interview, care_options, Work_Interest	Growing_Stress, Mood_Swings, Social_Weakness	-

Table 6: Performance of LLMs on multiple-choice prediction tasks in the PAS scenario. Accuracy (↑) is reported as avg. ± std. across repeated runs. Bold values indicate the best performance.

Dataset	Target Attributes	Zero-Shot Evaluation				Few-Shot Evaluation			
		GPT-3.5 Turbo	GPT-4 Turbo	LLaMA 3.0	LLaMA 3.1	GPT-3.5 Turbo	GPT-4 Turbo	LLaMA 3.0	LLaMA 3.1
ANES	meeting	$0.7898_{\pm 0.0650}$	$\mathbf{0.8440}_{\pm 0.0239}$	$0.8314_{\pm 0.0602}$	$0.7619_{\pm 0.0099}$	$0.8002_{\pm 0.0527}$	$\mathbf{0.8353}_{\pm 0.0223}$	$0.7803_{\pm 0.0829}$	$0.7388_{\pm 0.0536}$
	online	$0.7431_{\pm 0.0876}$	$\mathbf{0.8069}_{\pm 0.0618}$	$0.7120_{\pm 0.0262}$	$0.6980_{\pm 0.0296}$	$0.7334_{\pm 0.0523}$	$\mathbf{0.8059}_{\pm 0.0541}$	$0.7134_{\pm 0.0215}$	$0.6789_{\pm 0.0354}$
	moneyorg	$0.8233_{\pm 0.0267}$	$\mathbf{0.8566}_{\pm 0.0116}$	$0.8047_{\pm 0.0597}$	$0.7975_{\pm 0.0721}$	$0.8051_{\pm 0.0290}$	$\mathbf{0.8244}_{\pm 0.0422}$	$0.8122_{\pm 0.0101}$	$0.7887_{\pm 0.0584}$
GSS	strmgtsup	$0.5225_{\pm 0.0777}$	$\mathbf{0.5893}_{\pm 0.0064}$	$0.5563_{\pm 0.0826}$	$0.5793_{\pm 0.0064}$	$0.4625_{\pm 0.0356}$	$\mathbf{0.6016}_{\pm 0.0264}$	$0.5289_{\pm 0.1266}$	$0.4868_{\pm 0.1217}$
	allorglevel	$0.5950_{\pm 0.1940}$	$\mathbf{0.6732}_{\pm 0.0615}$	$0.5016_{\pm 0.0339}$	$0.4145_{\pm 0.0588}$	$0.5270_{\pm 0.0503}$	$\mathbf{0.6429}_{\pm 0.0433}$	$0.4577_{\pm 0.0523}$	$0.3389_{\pm 0.0338}$
BIS	vote	$0.1865_{\pm 0.0032}$	$0.2165_{\pm 0.0022}$	$0.2629_{\pm 0.0600}$	$\mathbf{0.2848}_{\pm 0.0217}$	$0.1465_{\pm 0.0164}$	$0.2169_{\pm 0.0045}$	$\mathbf{0.2261}_{\pm 0.0233}$	$0.2077_{\pm 0.0044}$
RECS	UGASHERE	$0.6208_{\pm 0.1116}$	$\mathbf{0.8598}_{\pm 0.0793}$	$0.7257_{\pm 0.0144}$	$0.6540_{\pm 0.0730}$	$0.6386_{\pm 0.0257}$	$\mathbf{0.9113}_{\pm 0.0165}$	$0.7483_{\pm 0.2132}$	$0.6742_{\pm 0.0623}$
Trell SMU	weekdays_trails_...	$0.6415_{\pm 0.0105}$	$\mathbf{0.6525}_{\pm 0.0050}$	$0.3865_{\pm 0.0138}$	$0.5028_{\pm 0.0134}$	$0.7228_{\pm 0.0357}$	$0.6601_{\pm 0.0057}$	$0.7099_{\pm 0.0051}$	$\mathbf{0.7309}_{\pm 0.0176}$
EmpS	WLB	$0.1849_{\pm 0.0114}$	$\mathbf{0.1959}_{\pm 0.0016}$	$0.1951_{\pm 0.0228}$	$0.1804_{\pm 0.0027}$	$0.1728_{\pm 0.0073}$	$0.1913_{\pm 0.0070}$	$\mathbf{0.1945}_{\pm 0.0147}$	$0.1898_{\pm 0.0026}$
	WorkEnv	$0.1943_{\pm 0.0036}$	$0.1949_{\pm 0.0029}$	$\mathbf{0.2080}_{\pm 0.0032}$	$0.1985_{\pm 0.0061}$	$0.1868_{\pm 0.0048}$	$0.1825_{\pm 0.0173}$	$0.2052_{\pm 0.0019}$	$\mathbf{0.2071}_{\pm 0.0050}$
	Workload	$0.2036_{\pm 0.0058}$	$0.1973_{\pm 0.0023}$	$0.2035_{\pm 0.0007}$	$\mathbf{0.2136}_{\pm 0.0191}$	$0.1940_{\pm 0.0106}$	$0.1952_{\pm 0.0125}$	$\mathbf{0.2001}_{\pm 0.0099}$	$0.1995_{\pm 0.0070}$
	Stress	$0.1278_{\pm 0.0824}$	$0.1268_{\pm 0.0015}$	$\mathbf{0.1847}_{\pm 0.0247}$	$0.1667_{\pm 0.0021}$	$0.1787_{\pm 0.0116}$	$0.1043_{\pm 0.0129}$	$0.1483_{\pm 0.0190}$	$\mathbf{0.2425}_{\pm 0.0028}$
	JobSatisfaction	$\mathbf{0.3852}_{\pm 0.0434}$	$0.3706_{\pm 0.0004}$	$0.3598_{\pm 0.0779}$	$0.3348_{\pm 0.0614}$	$\mathbf{0.4013}_{\pm 0.0118}$	$0.3848_{\pm 0.0029}$	$0.3252_{\pm 0.0696}$	$0.2327_{\pm 0.0049}$
NHTS	TRPTRANS	$\mathbf{0.7314}_{\pm 0.0077}$	$0.7231_{\pm 0.0003}$	$0.5732_{\pm 0.0126}$	$0.6507_{\pm 0.0453}$	$0.7246_{\pm 0.0322}$	$\mathbf{0.7300}_{\pm 0.0057}$	$0.6409_{\pm 0.0797}$	$0.5987_{\pm 0.0270}$
MxMH	Anxiety	$0.0650_{\pm 0.0036}$	$\mathbf{0.0957}_{\pm 0.0000}$	$0.0704_{\pm 0.0083}$	$0.0898_{\pm 0.0175}$	$\mathbf{0.1027}_{\pm 0.0332}$	$0.0888_{\pm 0.0065}$	$0.0695_{\pm 0.0104}$	$0.0771_{\pm 0.0105}$
	Depression	$0.0966_{\pm 0.0414}$	$\mathbf{0.1049}_{\pm 0.0010}$	$0.0892_{\pm 0.0144}$	$0.0972_{\pm 0.0167}$	$\mathbf{0.0977}_{\pm 0.0169}$	$0.0889_{\pm 0.0158}$	$0.0963_{\pm 0.0067}$	$0.0970_{\pm 0.0175}$
	Insomnia	$\mathbf{0.1040}_{\pm 0.0702}$	$0.0903_{\pm 0.0010}$	$0.0828_{\pm 0.0113}$	$0.0876_{\pm 0.0123}$	$\mathbf{0.1784}_{\pm 0.0618}$	$0.0968_{\pm 0.0056}$	$0.0897_{\pm 0.0090}$	$0.0709_{\pm 0.0080}$
	OCD	$\mathbf{0.2476}_{\pm 0.0367}$	$0.1174_{\pm 0.0052}$	$0.1179_{\pm 0.0177}$	$0.0986_{\pm 0.0137}$	$\mathbf{0.2379}_{\pm 0.0948}$	$0.1165_{\pm 0.0091}$	$0.1125_{\pm 0.0150}$	$0.0992_{\pm 0.0120}$
YPS	Action	$\mathbf{0.2388}_{\pm 0.0074}$	$0.2191_{\pm 0.0016}$	$0.2203_{\pm 0.0686}$	$0.2068_{\pm 0.0483}$	$0.2465_{\pm 0.0204}$	$\mathbf{0.2939}_{\pm 0.0681}$	$0.1602_{\pm 0.0001}$	$0.1073_{\pm 0.0081}$
	Documentary	$\mathbf{0.2952}_{\pm 0.0026}$	$0.2698_{\pm 0.0018}$	$0.2731_{\pm 0.0157}$	$0.2885_{\pm 0.0232}$	$\mathbf{0.2948}_{\pm 0.0119}$	$0.2758_{\pm 0.0303}$	$0.2809_{\pm 0.0186}$	$0.2092_{\pm 0.0823}$
	Thriller	$0.2462_{\pm 0.0093}$	$\mathbf{0.3482}_{\pm 0.0058}$	$0.3033_{\pm 0.0191}$	$0.2637_{\pm 0.0286}$	$\mathbf{0.3214}_{\pm 0.0213}$	$0.3132_{\pm 0.0493}$	$0.2592_{\pm 0.0268}$	$0.2078_{\pm 0.0171}$
	Comedy	$0.3115_{\pm 0.0104}$	$0.1865_{\pm 0.0059}$	$\mathbf{0.4366}_{\pm 0.0880}$	$0.3557_{\pm 0.0702}$	$0.3035_{\pm 0.0597}$	$0.3044_{\pm 0.0405}$	$\mathbf{0.4068}_{\pm 0.0243}$	$0.2716_{\pm 0.0596}$
MHD	Growing_Stress	$0.3231_{\pm 0.0034}$	$0.3454_{\pm 0.0088}$	$\mathbf{0.3461}_{\pm 0.0103}$	$0.3374_{\pm 0.0026}$	$0.3349_{\pm 0.0166}$	$0.3416_{\pm 0.0072}$	$\mathbf{0.3420}_{\pm 0.0083}$	$0.3369_{\pm 0.0018}$
	Mood_Swings	$0.3446_{\pm 0.0050}$	$0.3225_{\pm 0.0026}$	$0.3362_{\pm 0.0037}$	$\mathbf{0.3466}_{\pm 0.0078}$	$0.3322_{\pm 0.0161}$	$0.3185_{\pm 0.0103}$	$0.3428_{\pm 0.0114}$	$\mathbf{0.3482}_{\pm 0.0062}$
	Social_Weakness	$0.3454_{\pm 0.0089}$	$0.3445_{\pm 0.0021}$	$0.3348_{\pm 0.0063}$	$\mathbf{0.3461}_{\pm 0.0107}$	$0.3272_{\pm 0.0266}$	$0.3417_{\pm 0.0063}$	$0.3361_{\pm 0.0084}$	$\mathbf{0.3460}_{\pm 0.0108}$

Table 7: Detailed Performance of LLMs on numerical prediction tasks in the PAS scenario. Performance Score (↑) is reported as avg. ± std. across repeated runs. Bold values indicate the best performance.

Dataset	Target Attributes	Zero-Shot Evaluation				Few-Shot Evaluation			
		GPT-3.5 Turbo	GPT-4 Turbo	LLaMA 3.0	LLaMA 3.1	GPT-3.5 Turbo	GPT-4 Turbo	LLaMA 3.0	LLaMA 3.1
ANES	ftobama1	$\mathbf{0.5974}_{\pm 0.0218}$	$0.5888_{\pm 0.0163}$	$0.4745_{\pm 0.0813}$	$0.4660_{\pm 0.1026}$	$\mathbf{0.6208}_{\pm 0.0008}$	$0.5499_{\pm 0.0754}$	$0.4631_{\pm 0.0949}$	$0.4823_{\pm 0.0822}$
	ftbiden1	$\mathbf{0.5788}_{\pm 0.0376}$	$0.5378_{\pm 0.0540}$	$0.4909_{\pm 0.0666}$	$0.4823_{\pm 0.1231}$	$0.5666_{\pm 0.0148}$	$\mathbf{0.5688}_{\pm 0.0583}$	$0.4654_{\pm 0.0724}$	$0.4529_{\pm 0.0173}$
	fttrump1	$0.5444_{\pm 0.0994}$	$\mathbf{0.5461}_{\pm 0.0177}$	$0.4971_{\pm 0.0119}$	$0.4820_{\pm 0.0007}$	$0.4829_{\pm 0.0061}$	$\mathbf{0.5191}_{\pm 0.0144}$	$0.4654_{\pm 0.0053}$	$0.4697_{\pm 0.0006}$
GSS	nateduc	$0.7062_{\pm 0.0138}$	$\mathbf{0.7139}_{\pm 0.0041}$	$0.6983_{\pm 0.0221}$	$0.6798_{\pm 0.0089}$	$0.6988_{\pm 0.0060}$	$\mathbf{0.7034}_{\pm 0.0276}$	$0.6959_{\pm 0.0118}$	$0.6753_{\pm 0.0366}$
	natcrime	$0.6904_{\pm 0.0101}$	$\mathbf{0.7024}_{\pm 0.0216}$	$0.6997_{\pm 0.0175}$	$0.6838_{\pm 0.0035}$	$0.6545_{\pm 0.0112}$	$\mathbf{0.7073}_{\pm 0.0052}$	$0.6798_{\pm 0.0165}$	$0.6848_{\pm 0.0250}$
BIS	argumentsfor	$\mathbf{0.6204}_{\pm 0.0462}$	$0.5119_{\pm 0.0129}$	$0.5777_{\pm 0.0458}$	$0.5730_{\pm 0.0378}$	$\mathbf{0.6207}_{\pm 0.0417}$	$0.5299_{\pm 0.0337}$	$0.5640_{\pm 0.0486}$	$0.5622_{\pm 0.0476}$
RECS	KWH	$\mathbf{0.6554}_{\pm 0.0162}$	$0.5648_{\pm 0.0568}$	$0.3114_{\pm 0.0801}$	$0.2669_{\pm 0.0197}$	$0.5293_{\pm 0.1865}$	$\mathbf{0.5489}_{\pm 0.0920}$	$0.3088_{\pm 0.0874}$	$0.2365_{\pm 0.0226}$
ACS	hrs_work	$0.5482_{\pm 0.1910}$	$0.4997_{\pm 0.2572}$	$\mathbf{0.6461}_{\pm 0.1041}$	$0.5807_{\pm 0.1182}$	$0.5265_{\pm 0.1232}$	$0.3259_{\pm 0.0281}$	$\mathbf{0.6506}_{\pm 0.1043}$	$0.6201_{\pm 0.0971}$
	income	$0.4835_{\pm 0.0655}$	$0.4447_{\pm 0.1146}$	$0.5650_{\pm 0.0925}$	$\mathbf{0.5725}_{\pm 0.1295}$	$0.4317_{\pm 0.0980}$	$0.4013_{\pm 0.0539}$	$0.5603_{\pm 0.1051}$	$\mathbf{0.5738}_{\pm 0.1252}$
Trell SMU	content_views	$0.3523_{\pm 0.0340}$	$0.4301_{\pm 0.0033}$	$\mathbf{0.5612}_{\pm 0.0003}$	$0.5361_{\pm 0.0064}$	$0.4274_{\pm 0.0499}$	$0.5197_{\pm 0.0670}$	$\mathbf{0.5462}_{\pm 0.0189}$	$0.4685_{\pm 0.0147}$
NHTS	TRPMILES	$0.5969_{\pm 0.0027}$	$0.4403_{\pm 0.1400}$	$0.5983_{\pm 0.1516}$	$\mathbf{0.6851}_{\pm 0.0481}$	$0.5507_{\pm 0.1309}$	$0.4194_{\pm 0.2138}$	$\mathbf{0.6435}_{\pm 0.1228}$	$0.5615_{\pm 0.0715}$
	TRVLCMIN	$0.2050_{\pm 0.0104}$	$0.1847_{\pm 0.0283}$	$0.3871_{\pm 0.2022}$	$\mathbf{0.5307}_{\pm 0.1590}$	$0.1629_{\pm 0.0684}$	$0.2011_{\pm 0.0860}$	$0.3381_{\pm 0.0887}$	$\mathbf{0.5982}_{\pm 0.0805}$

Table 8: Attribute selection details for prediction tasks in the FAS scenario. A subset of variables was selected as target attributes based on domain expertise, with input attributes chosen to ensure scientific rigor. Attributes with an underscore in the "Input Attributes" column are optional, excluded in the zero-context paradigm and included in the context-enhanced paradigm. Italicized terms refer to textual descriptions rather than specific attributes.

Dataset	**Simulation Objectives**	**Input Attributes**	**Target Attributes**
GSS	Public Opinions	*background*, sample size, batch size, distribution (military, crime, education)	natenvir, natheal, natcity
Trell SMU	Media Usage Habit	*background*, sample size, batch size, distribution (age group, city tier, duration, creation, averageCompletion)	content_views, weekends_trails_watched_per_day, weekdays_trails_watched_per_day
YPS	Music & Movie Preferences	*background*, sample size, batch size, distribution (Music, Slow songs or fast songs, Dance, Folk, Country, Classical music, Musical, Pop, Rock, Metal or Hardrock, Punk, Hiphop, Rap, Reggae, Ska, Swing, Jazz, Rock n roll, Alternative, Latino, Techno, Trance, Opera, Movies, Horror, Romantic, Sci-fi, War, Fantasy/Fairy tales, Animated)	Action, Documentary, Thriller, Comedy
RECS	Household Fuel Usage	*background*, sample size, batch size, distribution (Gender (HHSEX), Age Range, Employment Status, State (State-Postal))	KWH, TOTALDOL

Table 9: Detailed Performance of LLMs on prediction tasks in the FAS scenario. Performance Score (↑) is reported for each setting. Bold values indicate the best performance within each zero-context or context-enhanced generation.

Dataset	**Target Attribute**	**Zero-Context Generation**				**Context-Enhanced Generation**			
		GPT-3.5 Turbo	GPT-4 Turbo	LLaMA 3.0	LLaMA 3.1	GPT-3.5 Turbo	GPT-4 Turbo	LLaMA 3.0	LLaMA 3.1
GSS	natenvir	**0.6555**	0.6495	0.6449	0.6487	0.6504	**0.7005**	0.6649	0.6812
	natheal	0.6526	0.6517	0.6477	**0.6533**	0.6683	**0.6955**	0.6635	0.6777
	natcity	0.6388	0.6457	**0.6596**	0.6540	0.6520	**0.6624**	0.6430	0.6589
Trell SMU	content_views	0.2992	**0.3455**	0.2772	0.3330	0.4160	**0.4276**	0.3219	0.3066
	weekends_trails_...	0.2848	0.2822	0.1823	**0.2921**	**0.2797**	0.2732	0.2025	0.2497
	weekdays_trails_...	0.3669	**0.3683**	0.2463	0.2724	0.3272	**0.3844**	0.2426	0.2814
YPS	Action	0.6525	**0.6968**	0.6478	0.6368	0.6526	**0.6968**	0.6478	0.6368
	Documentary	0.6472	**0.6968**	0.6599	0.6542	0.6472	**0.6968**	0.6599	0.6542
	Thriller	0.6650	**0.7045**	0.6385	0.6407	0.6650	**0.7045**	0.6385	0.6407
	Comedy	0.7409	**0.7859**	0.6815	0.6837	0.7409	**0.7859**	0.6815	0.6837
RECS	KWH	0.6208	**0.6267**	0.5484	0.3753	0.5976	**0.6302**	0.5763	0.5113
	TOTALDOL	0.6460	**0.6501**	0.5709	0.3900	0.6209	**0.6504**	0.6003	0.5946

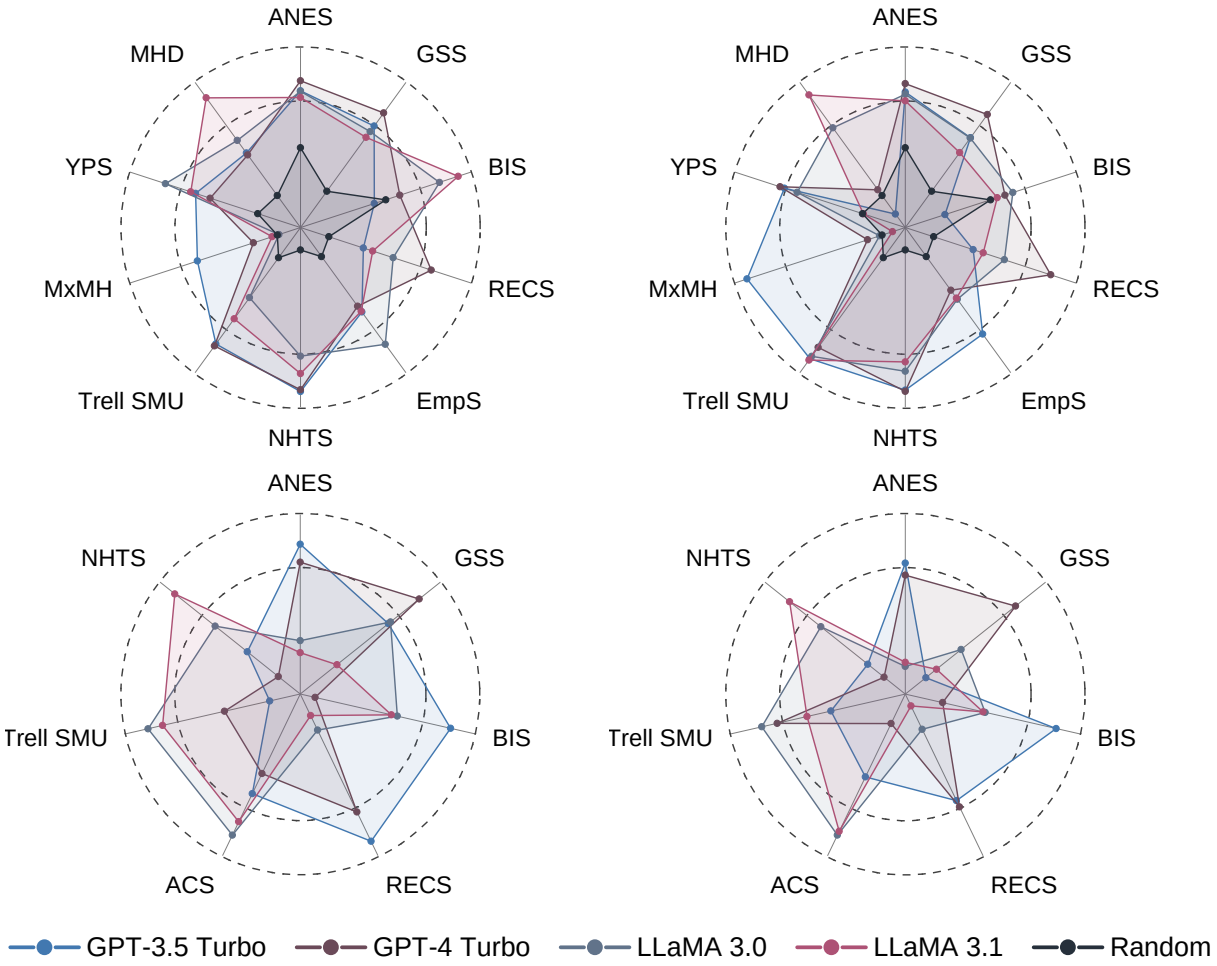


Figure 8: Radar plots illustrating model performance on the PAS scenario across datasets (each axis is individually scaled per dataset to enhance the visibility of relative differences). The top row shows multiple-choice tasks, and the bottom row shows numerical prediction, under zero-shot evaluation (left) and few-shot evaluation (right). Higher values indicate better performance.

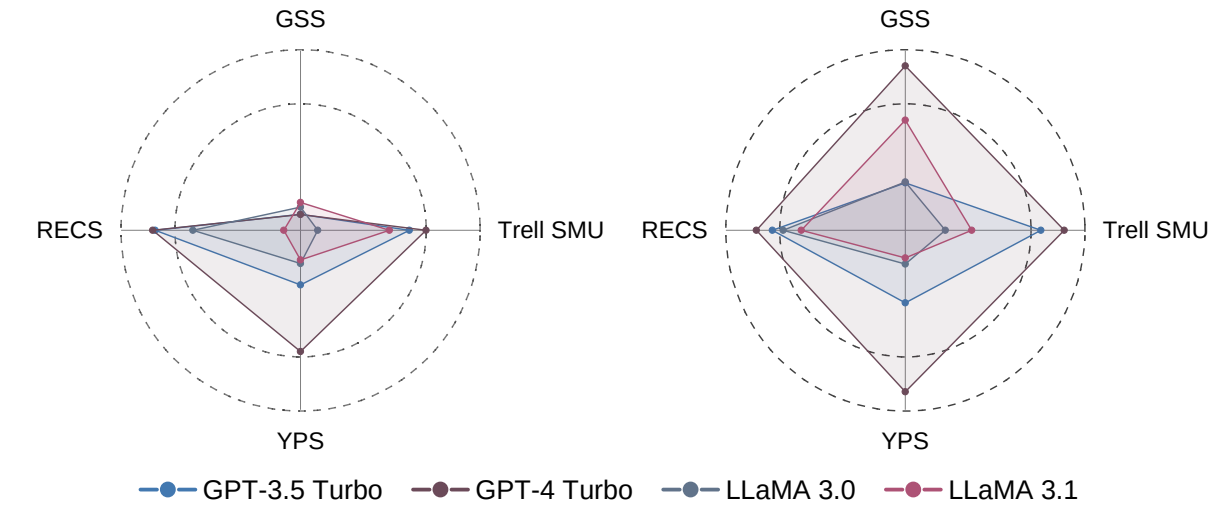


Figure 9: Radar plots illustrating model performance on the FAS scenario across datasets (each axis is individually scaled per dataset to enhance the visibility of relative differences). Results are shown under zero-context generation (left) and context-enhanced generation (right). Higher values indicate better performance.

Partial Attribute Simulation

Background Description

The survey content was shaped to a significant degree by ideas offered by the ANES user community through public solicitations and by members of the ANES National Advisory Board. Additional questions were included to capture changes in the political environment and recent developments in social science research. The questionnaire includes questions about voting behavior, candidate traits, political engagement, ideological orientations, racial identity and stereotyping, and many topical issues, including #MeToo, immigration, impeachment, and the coronavirus pandemic.

Respondent Profiles

This person is 40 years old, a homeowner, and has an income between $5,000 and $9,999. He has a voting tendency to select someone to vote for, but their summarized participation score is 0.

Conditions

This person's participation in political activities is as follows:

- *Has not attended a meeting to discuss political or social concerns.*
- *Has not donated to an organization focused on a political or social issue.*
- *Has not joined a protest march, rally, or demonstration.*
- *Has not posted messages or comments online about political issues or campaigns.*
- *Has not tried to persuade anyone to vote in any direction.*
- *Has not worn a campaign button, placed a sticker on their car, or displayed a sign in their window or yard.*
- *Has a cold or unfavorable feeling (score 15.0, in the range of [0,20]) towards Mike Pence and Andrew Yang.*
- *Has a cold or unfavorable feeling (score 0.0, in the range of [0,20]) towards Nancy Pelosi, Marco Rubio, Alexandria Ocasio-Cortez, and Nikki Haley.*

Task Demands

The data come from the American National Election Studies 2020 (ANES2020). As a sociologist and political scientist, you need to analyze the public opinion of citizens in this dataset and predict their possible election attitudes. Based on the information provided, can you help decide how this person would rate Clarence Thomas?

Few-Shot Examples (Zero-Shot Not Used)

Here are some examples and corresponding answers:

Example 1:

- *This person is 69 years old.*
- *This person is a homeowner.*
- *This person has an income between $10,000 and $14,999.*
- *This person probably would not vote.*
- *This person has a summarized participation score of 0.*
- *This person has a neutral feeling (score 60.0, in the range of [40, 60]) towards Mike Pence, Andrew Yang, Dr. Anthony Fauci, Joe Biden, and Donald Trump.*
- ***The answer is 60.***

Example 2: *...*
Example 3: *...*

Answer Format

Your response should consist of just one number between [0, 100] to reflect the person's attitude, without any additional text, explanation, or even a space character. Here is an example of the required response format that you should follow:

- *If you think this person has a slightly warm or favorable feeling, you should respond with JUST an integer number, such as 62.*
- *For more or less favorable feelings, the number may be larger or smaller.*
- ***So your response should be like: 62.***

Figure 10: A representative prompt utilized in zero-shot and few-shot evaluation conducted on the ANES dataset within the PAS scenario.

Full Attribute Simulation

Background Description

The Residential Energy Consumption Survey (RECS), conducted by the U.S. Energy Information Administration (EIA), is a nationally representative study designed to collect detailed information on household energy usage, expenditures, and related demographics. Since its inception in 1978, RECS has played a vital role in providing insights into energy consumption patterns across various housing units in the United States. The 2015 RECS dataset, for instance, surveyed over 5,600 households, representing approximately 118.2 million primary residences. The survey encompasses data on energy sources such as electricity, natural gas, propane, and fuel oil, along with expenditure estimates for heating, cooling, and other end uses. This dataset has been instrumental in energy policy analysis, efficiency improvements, and forecasting future consumption trends, making it a cornerstone of energy research in the residential sector.

Zero-Context Generation

You are a data scientist and socioeconomic analyst. Your task is to generate a synthetic dataset simulating residential energy consumption and household characteristics based on general knowledge and logical assumptions. No prior distributional information or feature statistics will be provided. Use your expertise to model realistic energy usage patterns, household demographics, and expenditure data for a nationally representative set of housing units.

Prediction Question

Based on your understanding, prior knowledge, and general reasoning, predict the following for a household:

- What is the total electricity use *(KWH)* of this household in 2020?
- What is the total electricity cost *(DOLLAREL)* of this household in 2020?
- What is the total energy cost *(TOTALDOL)* of this household in 2020?

Sample Size, Batch Size, and Answer Format

Generate 20 rows of data in each batch, and repeat this process for 100 batches to simulate a comprehensive dataset. For each batch, provide the predictions directly in the following format: [*KWH*, *DOLLAREL*, *TOTALDOL*]. After completing a batch, explicitly indicate the batch number before moving to the next batch (e.g., "Batch 1 complete"). Continue this process until all 100 batches are generated. The format must be a JSON string representing a three-dimensional array. Also, make sure that it is an array of arrays with no objects, like in a spreadsheet. After generating, only show the data you generated without additional words. Remember, the records should closely reflect the RECS dataset.

Context-Enhanced Generation

You are a data scientist and socioeconomic analyst. I will provide you with the demographic distribution of a population, including the proportions of gender *(HHSEX)*, age groups *(HHAGE)*, employment status *(EMPLOYHH)*, and state location *(state_postal)*. Based on these distributions, your task is to generate a synthetic dataset that reflects the specified demographic structure. The generated dataset should include plausible and consistent combinations of features that align with the given proportions, simulating a realistic social survey dataset. Ensure that the dataset captures meaningful patterns and relationships between features commonly observed in real-world surveys.

Distribution Information

You may be provided with the following distribution data:

- *HHSEX (sex)*: [Female: 0.5408, Male: 0.4592]
- *HHAGE (age group)*: [18-30: 0.0868, 31-40: 0.1503, 41-50: 0.1496, 51-60: 0.1848, 61-70: 0.2087, 71-80: 0.1574, 81-90: 0.0623]
- *EMPLOYHH (employment status)*: [Employed full-time: 0.4706, Employed part-time: 0.0868, Not employed: 0.1166, Retired: 0.3261]
- *state_postal (state location)*: [Alabama: 0.0131, Alaska: 0.0168, Arkansas: 0.0145, California: 0.0623, Colorado: 0.0195, Connecticut: 0.0159, Delaware: 0.0077, ..., Wisconsin: 0.0193, Wyoming: 0.0103]

Prediction Question

Based on the provided distributions of gender *(HHSEX)*, age range *(HHAGE)*, employment status *(EMPLOYHH)*, and state location *(state_postal)*, predict the total electricity use *(KWH)*, the total electricity cost *(DOLLAREL)*, and the total energy cost *(TOTALDOL)* for each household in 2020.

Sample Size, Batch Size, and Answer Format

(Same as Zero-Context Generation)

Figure 11: A representative prompt utilized in zero-context and context-enhanced generation conducted on the RECS dataset within the FAS scenario.